

# GENESIS AI ENGINEERING HANDBOOK

## PART VI

### DATABASE SPECIFICATION

**Version:** 1.0

**Status:** Draft

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#### Chapter 1 — Purpose

The Genesis Database is the company's memory.

Every department reads from and writes to the database.

No department stores permanent business data internally.

The database is the **single source of truth**.

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#### Chapter 2 — Database Principles

The Genesis Database must be:

- Source of Truth
  - Versioned
  - Auditable
  - Provider Independent
  - Scalable
  - Extensible
  - Human Readable
  - AI Friendly
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#### Chapter 3 — Phase 1 Storage

Current Storage

Google Sheets

Google Drive

Purpose

Rapid development

Zero infrastructure cost

Easy founder review

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## Chapter 4 — Phase 2 Storage

Future Database

Supabase

Object Storage

Cloudflare R2

Workflow Storage

n8n

Caching

Redis (Future)

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## Chapter 5 — Database Domains

Genesis stores information in domains.

Research

Products

Creative

Marketing

Publishing

Analytics

Administration

Knowledge Base

# Chapter 6 — Master Tables

## Table 1

Businesses

Purpose

Every business Genesis creates.

Fields

BusinessID

BusinessName

Industry

Status

CreatedDate

Owner

CurrentPhase

Priority

Notes

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## Table 2

Brands

Fields

BrandID

BusinessID

BrandName

Mission

TargetAudience

BrandVoice

Status

Logo

ColorPalette

Typography

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### **Table 3**

Projects

Purpose

Every founder request.

Fields

ProjectID

BusinessID

BrandID

Title

Description

Department

Priority

CurrentStatus

CreatedDate

ApprovedBy

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## Chapter 7 — Research Tables

ResearchReport

TrendReport

CompetitorReport

CustomerPersona

MarketScore

RiskAssessment

OpportunityScore

Each report is versioned.

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ResearchReport Fields

ResearchID

ProjectID

Version

DemandScore

CompetitionScore

ProfitScore

BrandPotential

AICompatibility

Recommendation

CreatedBy

ApprovedBy

Timestamp

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## Chapter 8 — Product Tables

Products

ProductCollections

Pricing

Bundles

Lifecycle

Products Fields

ProductID

BrandID

CollectionID

Name

Category

Description

SellingPrice

EstimatedCost

EstimatedMargin

Status

Version

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## Chapter 9 — Creative Tables

CreativeBrief

DesignPrompt

DesignAsset

Mockup

BrandGuideline

DesignPrompt Fields

PromptID

ProductID

Style

Prompt

Version

ApprovalStatus

GeneratedDate

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## Chapter 10 — Marketing Tables

Campaign

InstagramPosts

FacebookPosts

SEOKeywords

EmailCampaigns

Campaign Fields

CampaignID

ProductID

Platform

Caption

Hashtags

CTA

Status

ScheduleDate

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## Chapter 11 — Publishing Tables

PublishingJobs

MarketplaceStatus

Provider

PublishingHistory

PublishingJob Fields

JobID

ProductID

Marketplace

Provider

Status

PublishedDate

ErrorLog

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## Chapter 12 — Analytics Tables

Sales

MarketingMetrics

Performance

KPIs

Recommendations

Performance Fields

PerformanceID

ProductID

Views



Clicks

CTR

Orders

Revenue

Profit

Recommendation

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## Chapter 13 — Knowledge Base

Genesis stores reusable knowledge.

Prompt Library

Templates

Research Templates

Brand Guidelines

SOP Library

Employee Handbook

Lessons Learned

Future AI employees learn from this domain.

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## Chapter 14 — Administration Tables

Departments

Employees

Roles

Permissions

Settings

Document Versions

## Chapter 15 — Relationships

Business



Brand



Project



Research



Products



Creative



Marketing



Publishing



Analytics

Every record links back to its originating business.

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## Chapter 16 — Naming Standards

Primary Keys

BusinessID

BrandID

ProjectID

ProductID

ResearchID

PromptID

CampaignID

JobID

ReportID

All IDs use UUIDs in Phase 2.

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## Chapter 17 — Status Standards

Every major entity uses standard statuses.

Draft

Pending

Approved

Rejected

In Progress

Completed

Published

Archived

Deleted

Custom statuses are prohibited.

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## Chapter 18 — Versioning

Everything is versioned.

Research Reports

Creative Prompts

Products

Marketing

Architecture

Handbook

Database

Genesis never overwrites history.

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## Chapter 19 — Google Sheets Mapping

Phase 1 Workbook

Workbook

Genesis Database

Sheets

01 Businesses

02 Brands

03 Projects

04 Research

05 Products

06 Creative

07 Marketing

08 Publishing

09 Analytics

10 Knowledge

11 Admin

12 Settings

Every sheet maps directly to a future Supabase table.

Migration requires no structural redesign.

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## Chapter 20 — Migration Strategy

Phase 1

Google Sheets

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Phase 2

Supabase

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Phase 3

Distributed Database

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Phase 4

Enterprise Data Platform

Each migration preserves:

IDs

Relationships

History

Versions

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## Chapter 21 — Database Rules

Every record has:

Created By

Created Date

Version

Status

Approval Status

Business Owner

No anonymous data.

No orphan records.

No duplicate business entities.

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## Chapter 22 — Future Expansion

Future tables include:

CRM

Finance

Inventory

Suppliers

Customers

Manufacturing

HR

Legal

Support Tickets

Knowledge Graph

The database is designed so these additions require new tables, not redesign of existing ones.

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# CTO Review

## Strengths

- ✓ Clean separation of business domains
  - ✓ Easy migration from Google Sheets to Supabase
  - ✓ Suitable for AI workflows
  - ✓ Supports audit history
  - ✓ Version controlled
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## Improvements Planned (v2)

- ER diagrams
- Column-level data types
- Foreign key specifications
- Indexing strategy
- Row-level security
- Backup strategy
- Multi-tenant schema
- AI memory tables
- Embedding/vector search support
- Cost optimization schema

These enhancements will be introduced before implementation begins.

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**Next Document:** Part VII – Sprint Specifications